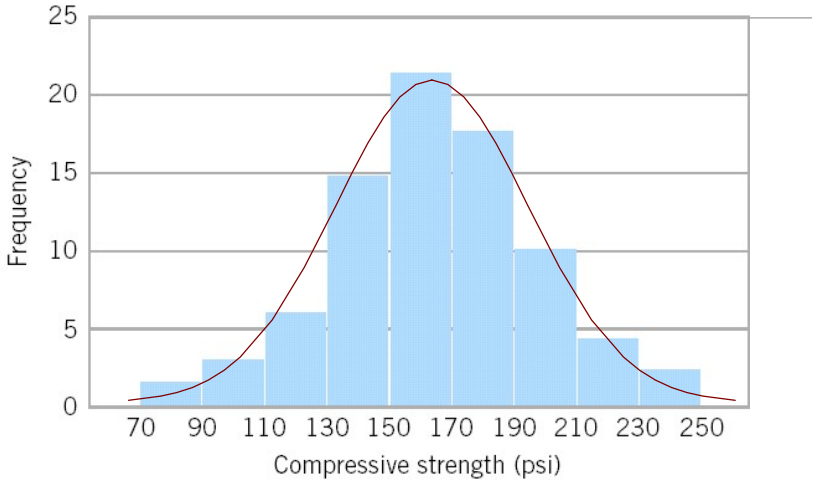
Probability Plots (Section 4.10)

- Probability plotting is a graphical method for determining whether sample data conform to a hypothesized distribution based on a subjective visual examination of the data.

1

The main idea



2

Quantiles of a distribution versus sample quantiles

Defn: Let p be a number between 0 and 1, the *100p*%th **population quantile** of the distribution of a continuous rv *X*, denoted by ς(p), is defined by:

$$p = F(\varsigma(p)) = \int_{-\infty}^{\varsigma(p)} f(y)dy$$

Defn: The 100*k*th **sample quantile** is a data value such that approximately 100*k*% of the observations are at or below this value and approximately 100(1 - *k*)% of them are above it.

- n ordered observations: j^{th} smallest → *100(j-0.5)/n)*th quantile

3

A probability plot

A plot of n pairs

*(j*th smallest observation, (*100(j-0.5)/n)*th quantile of the distribution)

on a two-dimensional coordinate system is called a probability plot.

If this plot is a reasonably straight line, then you may conclude that the sample came from the distribution that we used to find quantiles.

We will always use a software to plot these, so we skip the details of computing population and sample quantiles. We will learn in the lab.

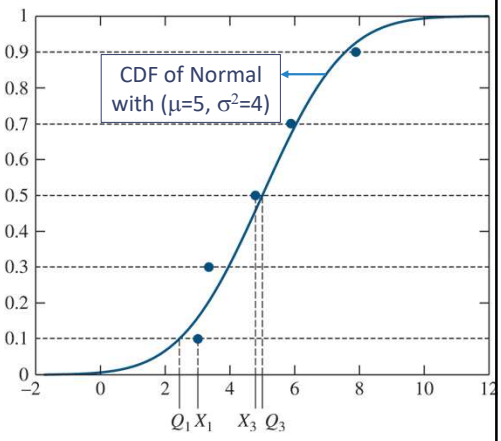
4

Example

3.01, 3.35, 4.79, 5.96, 7.89

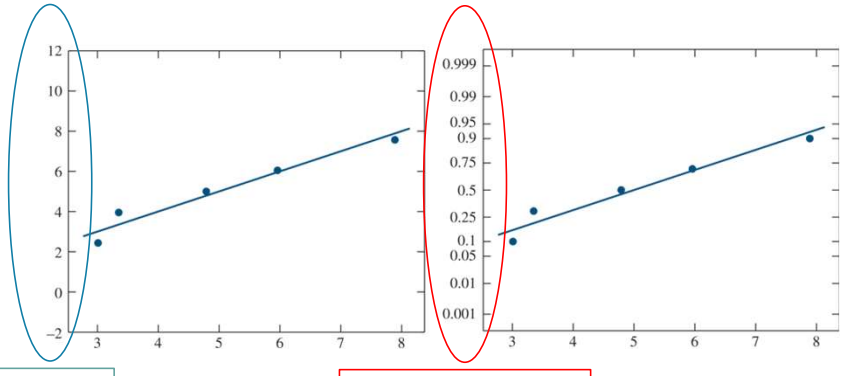
$\bar{x} = 5$ and $s = 2$

i	x_i	$(i - 0.5)/5$
1	3.01	0.1
2	3.35	0.3
3	4.79	0.5
4	5.96	0.7
5	7.89	0.9



5

Example

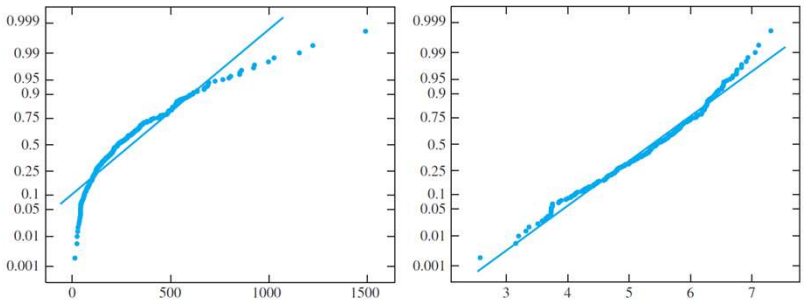


Percentiles of N(5,4)

Percent corresponding to N(5,4)

6

Normal Probability Plots

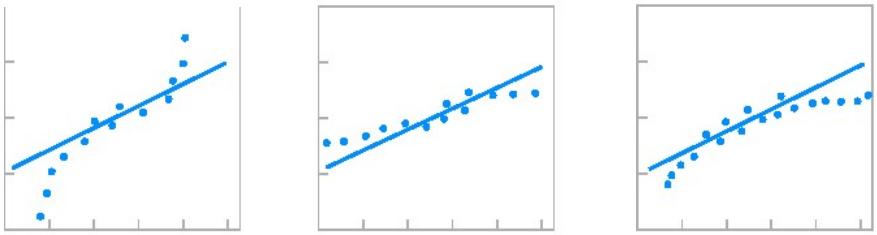


Not normal

Close to normal

7

Normal Probability Plots



Symmetric and light-tailed distribution

Symmetric and heavy-tailed distribution

A distribution with positive (or right) skew

8

Central Limit Theorem

- Central Limit Theorem applet with different distributions.
- You will observe this in Python simulations in the lab exercises and assignments.

9

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Example for CLT

Suppose that a random variable X has a uniform distribution with parameters (10,70).

What is the mean and standard deviation of X? $\mu = 40,\ \sigma = 17.32$

What is the approximate distribution of the sample mean of a random variable of size n=40?

Can we find its parameters? $\mu_{\bar{X}} = 40,\ \sigma_{\bar{X}} = 2.74$

What is the probability that this sample mean is larger than 44?

10

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Example for CLT

Suppose that a random variable X has a uniform distribution with parameters (10,70).

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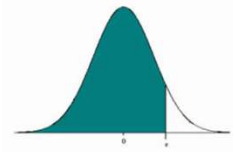
What is the approximate distribution of the sample mean of a random variable of size n=40?

Can we find its parameters?

What is the probability that this sample mean is larger than 44?

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Table of Standard Normal Probabilities for Positive Z-scores



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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Example 23

If a fair coin is tossed 100 times, use the normal curve to approximate the probability that the number of heads is between 45 and 55 *inclusive*.

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Normal approximation to Binomial
(DeMoivre-Laplace theorem)

We can have a slightly refined approximation by a small correction to account for the discreteness of the Binomial distribution.

P($X \leq x$) ≈ area under the normal curve to the left of $x + 0.5$

or equivalently:

$$P(k \leq X \leq l) \cong \Phi\left(\frac{l + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

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Normal Approximation to the Poisson

If $X \sim \text{Poisson}(\lambda)$, where $\lambda > 10$, then $X \sim N(\lambda, \lambda^2)$.

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Continuity Correction

The binomial distribution is discrete, while the normal distribution is continuous.

The continuity correction is an adjustment, made when approximating a discrete distribution with a continuous one, that can improve the accuracy of the approximation.

If you want to include the endpoints in your probability calculation, then extend each endpoint by 0.5. Then proceed with the calculation.

If you want exclude the endpoints in your probability calculation, then include 0.5 less from each endpoint in the calculation.

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